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Identifying habitat connectivity gaps using wildlife roadkill patterns in Ventura County, California

Jungbin Yoon[1], Chul Jin Yoon[2]

[1] The Thacher School, Ojai, California
[2] Parent and Independent Research Supervisor

Student Authors
Jungbin Yoon, High School

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OVERVIEW: *Habitat fragmentation caused by road networks can disrupt wildlife movement and increase mortality from vehicle collisions. In this study, spatial analysis of wildlife roadkill observations, land cover data, and transportation networks was used to examine patterns of habitat connectivity in Ventura County, California. The results identify areas where conservation efforts could improve wildlife movement and reduce wildlife–vehicle collisions.*

SUMMARY

Habitat fragmentation caused by expanding transportation networks is an increasing concern for wildlife conservation. Roads can restrict animal movement and increase mortality through vehicle collisions. Understanding where wildlife interactions with roads occur most frequently can help identify areas where habitat connectivity may be disrupted and where conservation efforts could be most effective. This study examined spatial patterns of wildlife road mortality and landscape connectivity in Ventura County, California. We hypothesized that wildlife roadkill observations would occur disproportionately close to major road networks and in areas where natural habitat intersects with transportation infrastructure. To investigate this question, we analyzed publicly available spatial datasets using geographic information systems and statistical tools in R. We combined land cover data, protected area boundaries, road network data, and wildlife roadkill observations to examine how landscape features influence wildlife–road interactions. The results suggested that roadkill observations occurred closer to roads than randomly distributed locations,

although this relationship may be partially influenced by reporting bias associated with citizen-science observations. These patterns suggest that roads may act as important barriers to wildlife movement while also concentrating areas of wildlife mortality. Identifying these spatial patterns can help conservation planning by highlighting areas where improving habitat connectivity or implementing mitigation strategies may reduce wildlife mortality to support more effective wildlife management.

INTRODUCTION

Habitat fragmentation caused by human development is widely recognized as one of the major threats to biodiversity (1). Expanding road networks, urban growth, and land conversion alter natural landscapes and disrupt ecological processes that support wildlife populations. Research in road ecology has shown that transportation infrastructure can produce a wide range of environmental impacts, including habitat loss, barriers to animal movement, and increased wildlife mortality (1). In a foundational study of road ecology, Forman and Alexander describe how roads influence ecological systems far beyond the roadway itself by reshaping wildlife movement patterns and altering landscape structure (1). These impacts can accumulate across large areas as road networks expand, making it increasingly difficult for animals to move safely between habitat patches.

The ecological influence of roads also extends beyond the physical roadway. Later work examining suburban highways found that transportation infrastructure can affect surrounding ecosystems across substantial distances, creating what has been described as a “road-effect zone,” where wildlife distribution and behavior change near roads (2). These zones may reduce habitat quality, alter animal movement patterns, and increase the likelihood of wildlife-vehicle collisions. Wildlife mortality is one of the most visible consequences of expanding road networks, and studies show that vehicle collisions can measurably affect wildlife populations (3). A large review examining road mortality across mammalian species found that wildlife–vehicle collisions can influence population dynamics by reducing survival, lowering population abundance, and increasing the risk of local population declines, particularly in landscapes with dense transportation infrastructure (3).

Beyond direct mortality, roads and urban development can fragment habitats and reduce ecological connectivity. Ecological connectivity refers to the ability of organisms to move across landscapes, which is essential for dispersal, gene flow, and long-term population stability (4, 5). Transportation planning has been found to be an important factor influencing habitat fragmentation, as road expansion can greatly affect wildlife movement and habitat connectivity if mitigation measures are not incorporated into infrastructure design (9). Studies on landscape connectivity emphasize that maintaining connections between habitat patches is critical for conserving biodiversity in human-dominated environments (4). When natural habitats become fragmented by roads or development, wildlife populations may become isolated within smaller patches, increasing the risk of local population decline.

Because of these challenges, conservation researchers have increasingly focused on identifying habitat corridors that allow animals to move between fragmented habitats. Work by

Beier and Noss highlighted the importance of habitat corridors for supporting wildlife movement between isolated habitat patches (5). Recent research analyzing wildlife movement patterns across landscapes has further emphasized the importance of locating key areas where animals are most likely to travel (6, 8). Studies examining animal movement data have shown that spatial patterns of wildlife activity can help find critical landscape features that facilitate connectivity between habitats (6). Advances in geographic information systems (GIS) and spatial analysis have made it possible to examine patterns of wildlife mortality and landscape structure using publicly available datasets. By combining land cover information, road network data, and wildlife observations, researchers can detect spatial patterns that reveal how wildlife interacts with human infrastructure.

The California Essential Habitat Connectivity Project identified key habitat linkages throughout the state and emphasized the importance of protecting movement corridors to maintain ecological connectivity across fragmented landscapes (8). Understanding wildlife movement is particularly important in southern California, where urban development and transportation infrastructure intersect with biodiverse natural landscapes. Research in southern California has shown that preserving the connectivity between habitat patches is crucial for supporting wildlife populations (7, 8). Ventura County, California contains a mixture of urban areas, agricultural land, and protected natural habitats (10, 11). Although several protected lands exist within the region, wildlife movement between these areas may be limited by road networks and other forms of development. Identifying spatial patterns of wildlife mortality and examining how these patterns relate to landscape features can provide insight into where habitat connectivity may be disrupted.

In this study, publicly available spatial datasets were used to examine patterns of wildlife mortality and landscape connectivity in Ventura County. Roadkill observations were analyzed alongside land cover and transportation network data to identify areas where wildlife–vehicle collisions occur most frequently. We hypothesized that wildlife mortality observations would occur significantly closer to roads than randomly distributed locations and that clusters of roadkill observations would highlight areas where habitat connectivity is most disrupted. Our analysis found that wildlife roadkill observations occurred significantly closer to roads than locations distributed by random and formed spatial clusters along major road networks. These findings suggest that transportation infrastructure may act as both a barrier to wildlife movement and a source of concentrated mortality, highlighting locations where improving habitat connectivity or implementing mitigation strategies could facilitate more effective conservation planning.

RESULTS

To evaluate whether wildlife roadkill observations identify habitat connectivity gaps in Ventura County, spatial analyses were conducted to examine roadkill hotspots, habitat suitability, and relationships between roadkill locations and transportation infrastructure.

Spatial distribution of roadkill observations

To examine where wildlife mortality events occurred across Ventura County, 13 recorded roadkill observations from the iNaturalist dataset collected between June 2020 and July 2024 were mapped relative to the regional road network (**Figure 1**). The mapped observations showed that roadkill events were not evenly distributed across the study area. Most observations were concentrated in the southern portion of Ventura County, while relatively few observations were recorded in the northern portion of the county (**Figure 1**). Several observations occurred near the Ventura-Oxnard urban area and adjacent major roadways (**Figure 1**).

Land cover Distribution

Land cover classification was used to provide descriptive landscape context for interpreting the spatial distribution of roadkill observations across Ventura County. Land cover data from the National Land Cover Dataset (NLCD 2021) were grouped into major habitat categories including forest, shrubland, grassland, agriculture, and urban development. The results showed that shrubland and grassland were the dominant land cover types across most of the county, particularly in the northern and interior mountainous regions (**Figure 2**). Forested areas occurred primarily at higher elevations, while agricultural and urban land cover were concentrated in the southern portion of the county. Urban land cover formed a large continuous area near the Ventura-Oxnard region where road networks are most dense.

Roadkill hotspot analysis

To identify areas where wildlife mortality events were spatially concentrated, a kernel density estimation (KDE) analysis was conducted using the roadkill observation points. This method highlights clusters of observations by estimating the density of points across the landscape. Given the limited sample size (n = 13), the KDE analysis was used as an exploratory visualization of areas where roadkill observations appeared to be concentrated rather than as definitive evidence of statistically significant hotspots. The KDE surface suggested several localized areas of relatively higher observation density, primarily in the southern and central portions of Ventura County (**Figure 3**). Areas with the highest densities were located near major road corridors and developed regions of the county. In contrast, northern areas with fewer roads showed very low-density values.

Habitat corridor suitability

A corridor suitability surface was generated to estimate areas that may support wildlife movement across Ventura County. This analysis combined land cover data, road networks, and roadkill observations to estimate how suitable different parts of the landscape may be for wildlife movement. Areas with higher suitability values were generally located in the northern portion of the county where natural habitats remain more continuous, and road density is lower (**Figure 4**). Lower suitability values were observed in the southern portion of the county where urban development and transportation infrastructure are more extensive. Several areas of moderate suitability occurred near clusters of roadkill observations.

Distance of roadkill observations from roads

To evaluate how closely wildlife mortality events occurred relative to roads, the distance from each roadkill observation to the nearest road was calculated. Distances ranged from values near 0 meters to more than 20,000 meters, although a small number of observations represented unusually large distances from mapped roads (**Figure 5**). Most observations occurred within the first several thousand meters of a road, and the distribution of distances was strongly skewed toward smaller values. Shorter distances (x–y meters) between roadkill observations and roads were far more common (n=x) than larger distances (x–y meters; n=x), indicating that many wildlife mortality events occurred close to transportation infrastructure (**Figure 5**).

Comparison of roadkill and random locations

To determine whether roadkill observations occurred closer to roads than would be expected by chance, the nearest-road distances for the 13 roadkill observations were compared with those of 13 randomly generated points distributed within the Ventura County boundary. Roadkill observations were generally located closer to roads than the randomly generated points (Figure 6). The median distance from roadkill observations to the nearest road (2,256.65 m) was greater than the median distance observed for the randomly generated points (290.54 m; Figure 6). Roadkill observations also exhibited a wider range of distances (approximately 0–20,300 m) from roads than the randomly generated points (approximately 0–7,400 m).

DISCUSSION

Animals moving through fragmented environments may be forced to cross roads when traveling between habitat patches, increasing the likelihood of vehicle collisions. Previous research in road ecology has shown that transportation infrastructure can alter wildlife movement patterns and contribute to wildlife mortality across landscapes (1, 3). The results of this study suggest that wildlife–vehicle collisions in Ventura County are closely associated with patterns of human development and transportation infrastructure. Roadkill observations were concentrated primarily in the southern portion of the county, where road networks and urban development are

most extensive. In contrast, the northern portion of the county, which contains more continuous natural habitat and fewer roads, had relatively few recorded observations. This spatial pattern indicates that wildlife mortality events are more likely to occur where natural landscapes intersect with dense road systems. While the data in this study did not demonstrate a direct cause-and-effect relationship, the strong spatial association between roadkill observations and transportation corridors supports the hypothesis that road networks influence wildlife mortality patterns within the region.

The land cover analysis provided additional context for understanding these patterns. Ventura County contains extensive shrubland and grassland habitats that support a wide range of wildlife species, particularly in the mountainous regions of the county. However, the southern coastal plain has experienced substantial urban and agricultural development, which has altered the natural landscape and increased habitat fragmentation. When large areas of habitat are divided by roads and urban infrastructure, wildlife movement between habitat patches becomes more difficult and may require animals to cross roads during dispersal or daily movement (1,5). The overlap between areas of development and the locations of roadkill observations suggests that these boundaries between natural habitat and human infrastructure may represent locations where wildlife encounters transportation networks most frequently. This does not necessarily mean that development directly causes wildlife mortality, but it indicates that fragmented landscapes may increase the probability that animals will interact with roads while moving through the environment.

The hotspot analysis further highlighted the uneven distribution of wildlife–vehicle collisions across the study area. Rather than being scattered randomly, roadkill observations formed several localized clusters near major road corridors. These hotspots may represent locations where wildlife regularly attempt to cross transportation routes while moving between nearby habitat patches. Certain road segments may therefore pose a higher risk to wildlife because of their position relative to surrounding habitats. Identifying these areas is important for conservation planning because mitigation strategies such as wildlife crossing structures, roadside fencing, and improved road design are often most effective when implemented at locations where wildlife crossing attempts are concentrated (9). Although this study did not directly measure animal movement behavior, the presence of clusters suggests that some parts of the road network may function as recurring crossing points for wildlife traveling across the landscape.

The corridor suitability analysis provided further insight into how landscape structure may influence wildlife movement. Areas with higher suitability values were primarily located in the northern portion of Ventura County, where natural habitats remain relatively intact and road

density is lower. These regions may provide more favorable conditions for wildlife movement because animals can travel through continuous habitat without encountering major barriers. In contrast, the southern portion of the county showed lower suitability values due to the presence of urban development and transportation infrastructure. Interestingly, several areas with moderate suitability were located near clusters of roadkill observations. This pattern suggests that wildlife may still attempt to move through fragmented landscapes when habitat patches remain nearby, even when roads create obstacles that increase the risk of mortality. Future studies could directly compare the identified corridor suitability patterns with existing wildlife corridor or movement barrier datasets to further validate and refine the habitat connectivity gaps.

The distance analysis provided additional evidence that wildlife mortality events tend to occur close to roads. Most roadkill observations were located within relatively short distances of transportation infrastructure, and the distribution of distances was heavily skewed toward smaller values. This finding is consistent with the expectation that wildlife–vehicle collisions occur when animals attempt to cross roads during movement between habitats. However, the analysis also revealed a few observations that were located much farther from mapped roads. These observations were considered outliers because their nearest-road distances were substantially greater than most of the roadkill observations, although no formal statistical outlier criterion was applied. These values may reflect spatial uncertainty in the iNaturalist dataset, post-collision animal movement before the observation was recorded, or coordinate mismatches between observation locations and mapped road networks. Because there was no objective basis for determining that these observations were erroneous, they were retained in the analysis. These outliers highlight a notable limitation of the dataset, and future studies with larger sample sizes could evaluate their influence through sensitivity analyses.

Comparing roadkill observations with randomly generated points gave a clearer picture of the relationship between wildlife mortality and road proximity. Random points were often located much farther from roads than the observed roadkill locations, indicating that wildlife mortality events were not distributed randomly across the landscape. Instead, the results suggest that collisions occur more frequently in areas where animals encounter transportation infrastructure during their movements. This comparison implies that proximity to roads may play an important role in the spatial distribution of wildlife mortality events in Ventura County.

Several limitations should be considered when interpreting the results of this study. The dataset used in the analysis contained a relatively small number of roadkill observations (n = 13), which may limit the statistical power of the spatial analyses. These patterns should be interpreted cautiously because a KDE may produce apparent clusters, even when only a small number of

observations are available, that may not persist with additional data. Therefore, the KDE results should be interpreted as an exploratory visualization of potential concentration patterns rather than definitive evidence of wildlife mortality hotspots. Future studies incorporating larger sample sizes and observations collected over longer time periods will improve the robustness of the analyses and allow stronger inferences regarding the spatial patterns of wildlife–vehicle collisions.

In addition, the roadkill observations used in this study were obtained from iNaturalist, a citizen science platform that relies on opportunistic public reporting rather than systematic surveys. Roadkill occurring near roads, populated areas, or locations that are frequently visited by people is more likely to be reported than roadkill occurring in remote areas, and some wildlife mortality events may go unrecorded altogether. This reporting bias may partially contribute to the finding that roadkill observations occurred closer to roads than randomly generated points. Therefore, the comparison with random locations should be interpreted as evidence of a spatial association rather than definitive proof that roads alone determine the distribution of wildlife mortality. Also, the randomly generated points were intended only as a simple baseline for comparison and do not represent a perfect null model because the observation dataset is influenced by human reporting patterns.

Finally, this study examined overall spatial patterns rather than species-specific behavior. Different wildlife species vary in movement behavior, habitat preferences, and home-range size, which may influence where wildlife–vehicle collisions occur. Because the dataset contained relatively few observations across multiple species, species-level analyses were beyond the scope of this study. Future studies using larger datasets could examine species-specific spatial patterns to better understand how ecological differences influence wildlife–vehicle collisions. Despite these limitations, the findings of this study provided useful insight into how landscape structure and transportation infrastructure interact to influence wildlife mortality.

In summary, this study examined the spatial distribution of wildlife road mortality in Ventura County and explored how these patterns relate to land cover, transportation networks, and potential wildlife movement corridors. The results support the hypothesis that roadkill events are concentrated near developed areas as well as major transportation corridors and tend to occur closer to roads than locations distributed by random chance. The data demonstrated that the concentration of roadkill observations near roads correlates with the idea that landscape fragmentation and human land use patterns can strongly affect where wildlife–vehicle collisions occur. Strategies such as wildlife crossing structures, improved road design, and habitat restoration could potentially reduce the risk of collisions in areas where wildlife movement and transportation infrastructure overlap. Understanding these spatial patterns is an important step

toward identifying areas where conservation efforts could improve habitat connectivity and reduce wildlife mortality across the region.

MATERIALS AND METHODS

Data sources

Several publicly available datasets were used for the spatial analysis. Land cover information was obtained from the Annual NLCD Land Cover 2021 dataset from the Multi-Resolution Land Characteristics (MRLC) Consortium (10). This dataset classifies land across the United States at a 30-meter resolution and was used to identify major land cover types such as forest, shrubland, grassland, agriculture, and urban areas. Data on protected lands were obtained from the California Protected Areas Database (CPAD) 2025b provided by the California Natural Resources Agency (11). These data were used to identify areas that are currently managed for conservation. Road network data were obtained from the 2021 TIGER/Line roads shapefiles from the U.S. Census Bureau, which provide the locations of highways and other transportation infrastructure (12). Wildlife roadkill observations from June 2020 to July 2024 were collected from iNaturalist, which contains georeferenced reports submitted by citizen scientists (13). After filtering observations to Ventura County and removing records that did not meet the study criteria, a total of 13 roadkill observations were included in the analyses. These observations were used as indicators of locations where wildlife frequently interacts with roads. Additional information on wildlife movement barriers was obtained from the Wildlife Movement Barriers dataset provided by the California Department of Fish and Wildlife (14). County boundary data from the U.S. Census Bureau were used to limit all datasets to the Ventura County study area (15).

Spatial processing

The spatial datasets were processed using R (version 4.5.1) and QGIS (version 4.0). First, all datasets were projected into the same coordinate reference system so that the layers would align correctly. The NLCD 2021 land cover raster was clipped to the Ventura County boundary and grouped into broader land cover categories to make the map easier to interpret. The iNaturalist roadkill observations were filtered so that only records located within Ventura County were included in the analysis, and the road dataset was clipped to the same study area.

Distance analysis

To examine the relationship between wildlife mortality and roads, the distance from each roadkill observation to the nearest road was calculated using spatial analysis tools in R. These distances were used to evaluate how closely wildlife mortality events occur relative to road infrastructure. For comparison, 13 random points (equal to the number of roadkill observations) were generated within the Ventura County boundary using an R function in the sf package, which

uniformly samples random locations within a polygon. The distance from each random point to the nearest road was then calculated using the same method. Comparing these two sets of distances provides a simple baseline for evaluating whether roadkill observations tend to occur closer to roads than randomly distributed locations.

Roadkill hotspot analysis

KDE was used to provide an exploratory visualization of areas where roadkill observations appeared to be spatially concentrated. KDE produces a continuous surface that estimates the relative density of observations across the landscape. A bandwidth of approximately 2 km was selected to emphasize broader regional patterns while reducing the influence of individual observations.

Habitat corridor suitability analysis

An exploratory corridor suitability surface was created by combining habitat suitability, distance from roads, and estimated roadkill density. National Land Cover Database classes were assigned suitability values ranging from 0 to 1. Forest was assigned a value of 1.00, shrubland 0.85, grassland 0.70, agricultural land 0.35, barren land 0.20, wetlands and water 0.10, and developed land 0.05; land-cover classes not otherwise specified were assigned a value of 0.30. Road suitability increased linearly with distance from the nearest mapped road, from 0 at the road to 1 at distances of 5 km or greater. The KDE surface was normalized by its maximum value and inverted so that areas with lower estimated roadkill density received higher suitability values. The final suitability score was calculated as 0.50 × habitat suitability + 0.30 × road suitability + 0.20 × inverse roadkill-density suitability and was constrained to values between 0 and 1. These weights were selected to give the greatest influence to habitat conditions while also incorporating road avoidance and observed roadkill patterns.

Spacial analysis

All spatial analysis was conducted using R with spatial libraries including *sf, terra*, and *ggplot2,* while final map visualization and layout were produced using QGIS (16–20).

ACKNOWLEDGMENTS

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Figures and Figure Titles/Captions

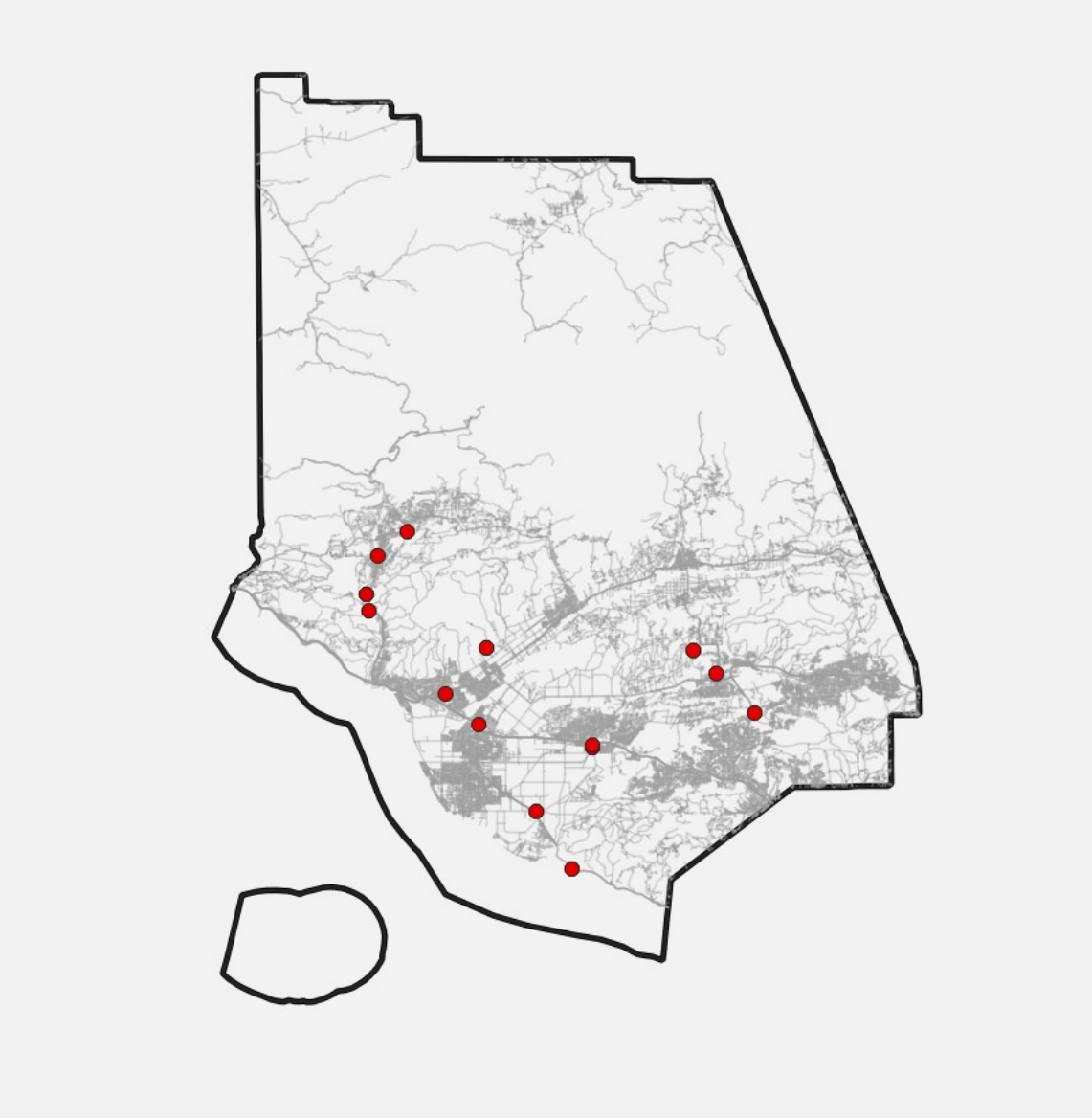


Figure 1: Study area and roadkill observations in Ventura County, California. The map shows the geographic boundary of Ventura County with the regional road network and the

recorded wildlife roadkill observations (n = 13; red) from iNaturalist from June 2020–July 2024. Gray lines represent roads, and black lines represent X. Road data were obtained from the U.S. Census Bureau TIGER/Line shapefiles, and observation data were compiled from research-grade wildlife reports.

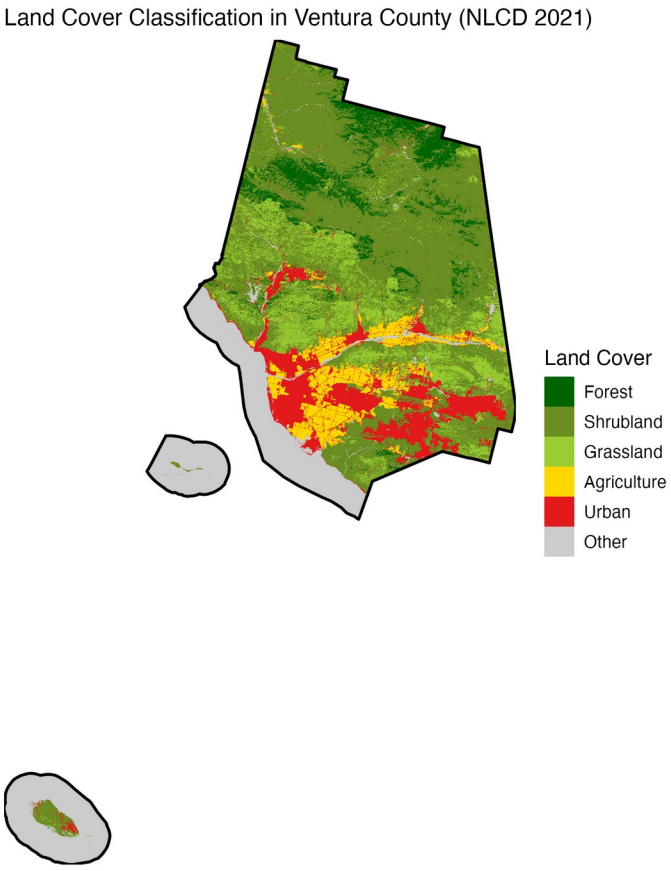


Figure 2: Land cover classification of Ventura County from the National Land Cover Database (NLCD 2021). The map displays major land cover categories across the study area used to represent habitat conditions relevant to wildlife movement. Land cover data were obtained from the NLCD 2021 and reclassified into forest, shrubland, grassland, agriculture, urban, and other categories for habitat suitability analysis.

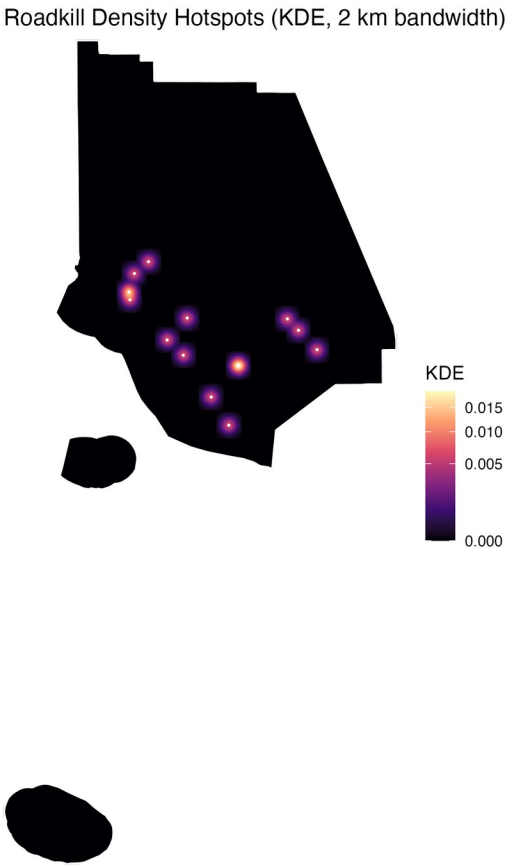


Figure 3: Exploratory kernel density estimation (KDE) of roadkill observations illustrating potential areas of higher observation density. The KDE surface summarized the spatial concentration of roadkill density hotspot observations (2 km bandwidth) by smoothing individual points into a continuous density surface, making localized clusters easier to visualize. Brighter colors indicate relatively higher estimated densities of roadkill observations, whereas darker colors indicate lower estimated densities. Because of the limited sample size (n = 13), the KDE analysis is presented as an exploratory visualization and should not be interpreted as statistically validated hotspots.

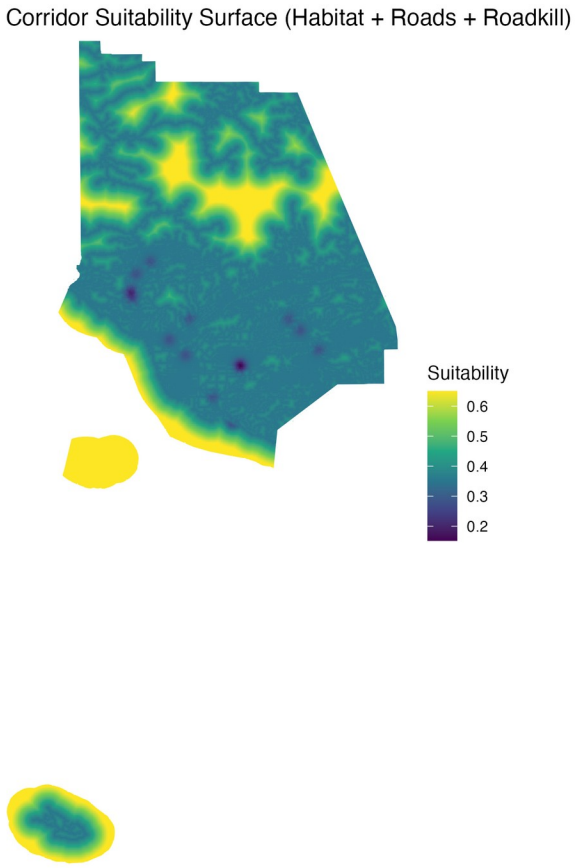


Figure 4: Exploratory corridor suitability surface combining habitat suitability, distance from roads, and inverse roadkill density. The surface was calculated using weights of 0.50 for habitat suitability, 0.30 for road-distance suitability, and 0.20 for inverse roadkill-density suitability. Warmer colors (yellow) indicate relatively higher suitability values under this exploratory weighting scheme, whereas cooler colors (blue-purple) indicate lower suitability values. The model was not independently validated and should not be interpreted as a predictive habitat-connectivity model.

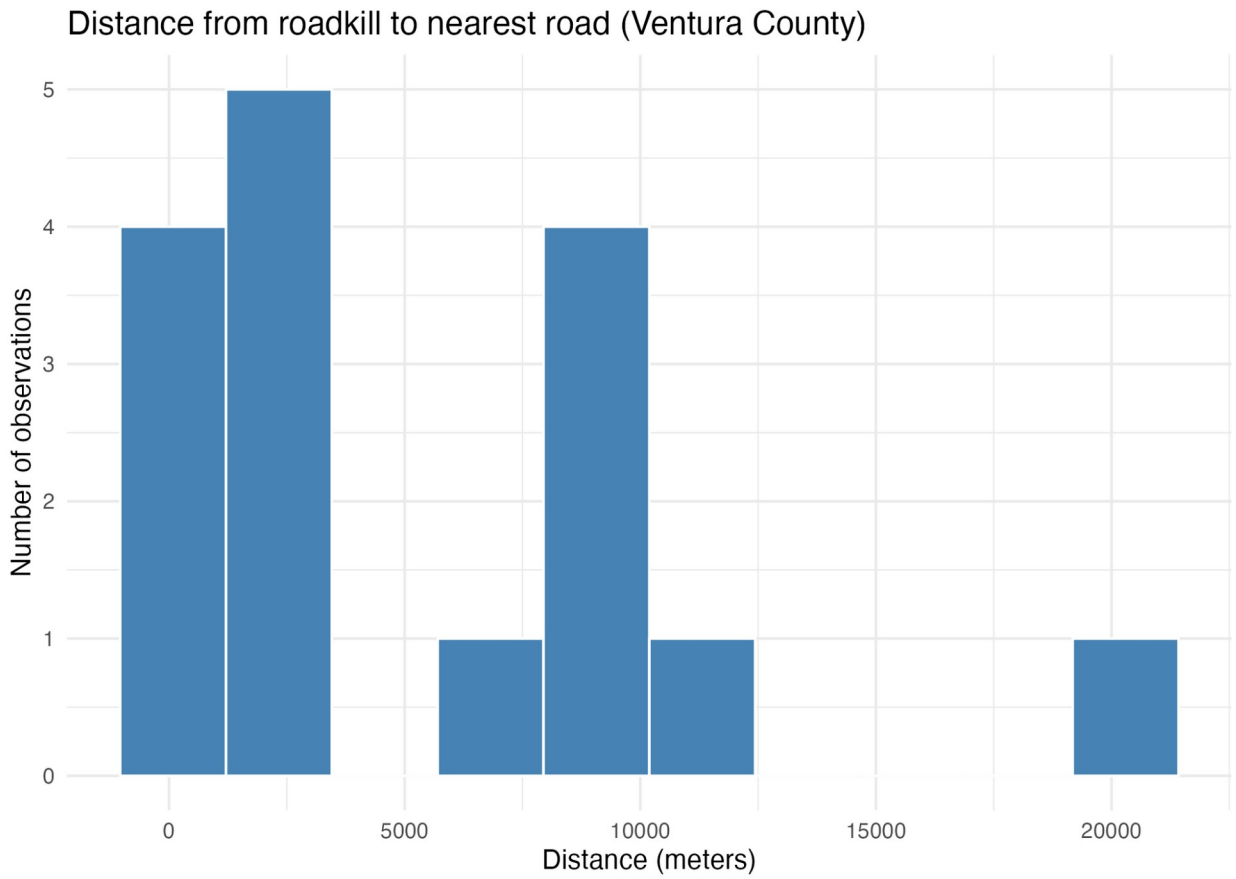


Figure 5: Distribution of distances from recorded roadkill observations to the nearest road in Ventura County. The histogram shows the frequency of distances between roadkill locations and the nearest road segment. Distances were calculated in meters using spatial nearest-neighbor analysis between observation points and the road network.

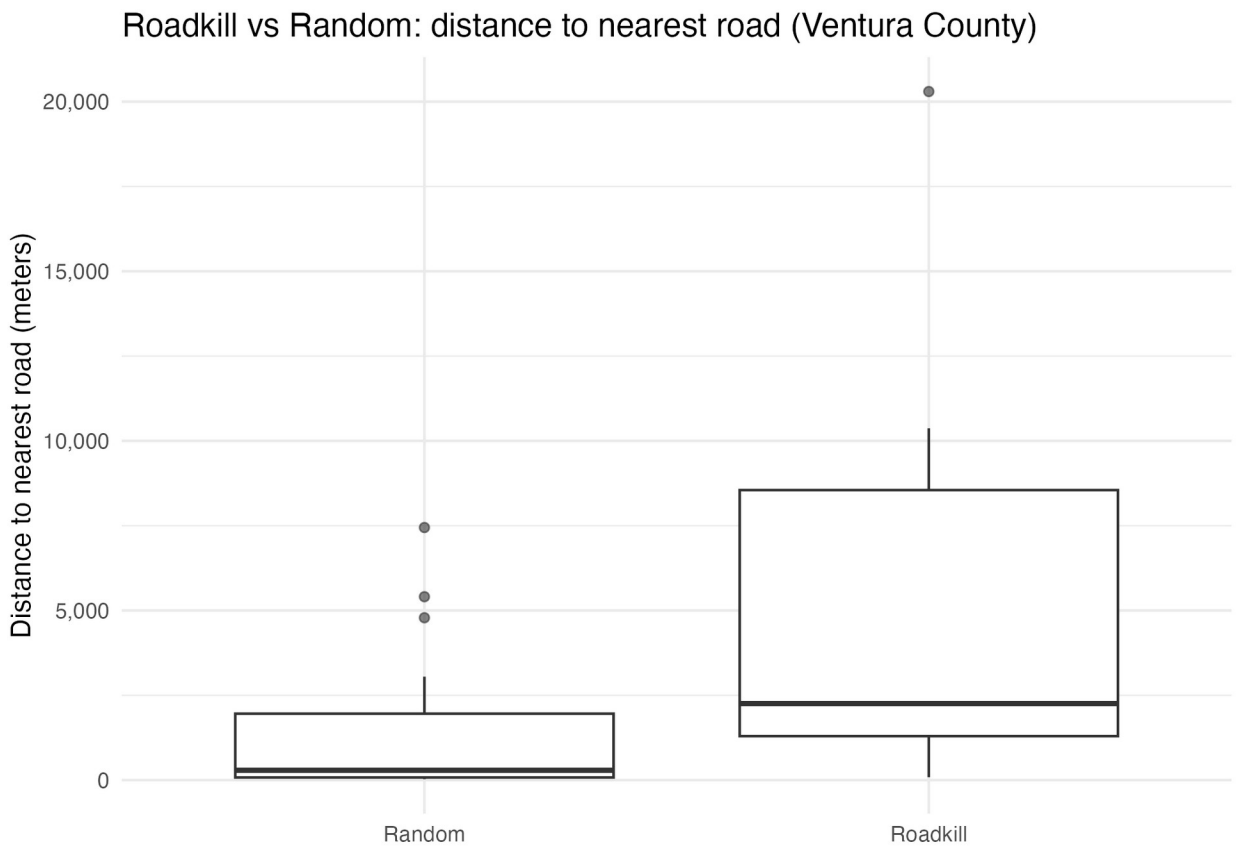


Figure 6: Comparison of distances to the nearest road for roadkill observations and randomly generated points. The boxplot compares how far roadkill observations occur from roads relative to randomly distributed points across the study area. The 13 random comparison points were generated by uniformly sampling locations within the Ventura County boundary using a function in the sf package. Distances were calculated in meters using nearest-neighbor distance analysis between each point and the road network. Roadkill and random distances were compared using a Wilcoxon rank-sum test.